# William Du

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## EDUCATION

**Carnegie Mellon University** — Mountain View, United States
Master of Science in Software Engineering — December 2023

**Queen's University** — Kingston, Canada
Bachelor of Applied Science in Computer Engineering — May 2022
Major GPA: 3.9/4.0

## PROFESSIONAL EXPERIENCE

**Lucid Motors** — Newark, United States
*Software Engineer* — Feb 2024 – Present

- Improved distributed automation framework performance by parallelizing execution across 45+ TCU devices, reducing full regression runtime by 85% and accelerating release validation cycles
- Developed and maintained 800+ automated tests using Python and Robot Framework, executed 1–3 times weekly per release, ensuring continuous validation of production software across NA, EU, and KSA
- Achieved an 80% reduction in system costs by creating a cost-effective design, reducing redundancies
- Designed and implemented long-duration stability and stress testing framework of CPU, memory, thermals, disk I/O, network, and service-level metrics, enabling proactive detection of performance degradation
- Managed large-scale Linux-based test farm infrastructure using Jenkins, GitLab CI, AWS, and MongoDB, supporting automated validation workflows across multiple engineering teams

**Restaurant Brands International** — Miami, United States
*Software Engineer Intern* — Jun 2023 – Aug 2023

- Deployed feature successfully to 52% of the total device population, benefiting 2.5M unique iPhone users
- Spearheaded the development of the iOS Live Activities feature and background fetching for Burger King
- Designed a custom package and patch to allow for proper linking and copying of files into native iOS app

**Queen's Predictive Geometallurgy and Geostatistics Lab** — Kingston, Canada
*Undergraduate Researcher* — May 2021 – Aug 2021

- Increased correlation coefficient of metallurgical models by 9% to an average accuracy of 79.6% while maintaining an average accuracy of 98% in mass recovery, validated on 1,690 observations
- Explored diverse machine learning configurations, including individual models per class, category-based grouping, and holistic modeling, achieving an R-squared of up to 0.999, optimizing model prediction
- Fine-tuned model parameters across 1,690 observations, leveraging advanced OLS regression techniques

## PROJECTS

**Manga Buddy** — Jan 2026 – Mar 2026

- Built a CV/NLP tool translating manga using a dual OCR pipeline (PaddleOCR + manga-ocr transformer)
- Implemented Union-Find clustering for text regions and collision-aware rendering for translated overlays
- Developed multi-threaded PyQt5 application with async ML inference to maintain a responsive UI

**Emergency Social Network** — Jun 2022 – Jan 2023

- Architected a real-time chatroom web application enabling seamless exchange of messages and images, supported by frameworks like ExpressJS and NodeJS, ensuring consistent and scalable performance
- Emphasized collaborative coding practices by integrating Continuous Integration, Development methodologies, and Test-Driven Development, ensuring both code quality and team alignment

## SKILLS

Python | Java | C/C++ | SQL | Linux | Scikit-learn | TensorFlow | OpenCV | React Native | TypeScript | Swift